

Ronald A. Ruffin

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PROFESSIONAL PROFILE

Senior Software Development Engineer with 20+ years of experience supporting, modernizing, and enhancing enterprise Java and .NET applications in secure, regulated environments. Extensive background in backend business logic, database systems, security modules, and automated testing within n-tier architectures.

Experienced in leading modernization initiatives, improving version control practices, implementing regression automation, and reducing technical debt in legacy systems.

Over the past several years, have integrated AI-assisted development workflows into both enterprise and personal projects — using large language models and GitHub Copilot to accelerate prototyping, refine architecture, strengthen unit testing, and compress feedback cycles while maintaining full engineering ownership and validation. My AI-augmented process is not “ask a chatbot for answers.” It is a disciplined loop: isolate the evidence, ask targeted questions, inspect the reasoning, test the result, and only then share the fix.

Known for disciplined testing, architectural clarity, and thoughtful system evolution rather than rapid but fragile delivery.

Built and deployed a publicly inspectable symbolic math application to demonstrate architectural rigor and AI-augmented SDLC practices beyond proprietary enterprise codebases.

TECHNICAL EXPERTISE

Languages:

Java (JSE/JEE), SQL/PL-SQL, JavaScript, Perl, C#, VB.NET, C/C++, JSP, Unix Shell

Frameworks & Platforms:

Spring Boot, Vaadin, PrimeFaces, JSF, JPA/Hibernate, Apache Shiro, Maven, Ant, n-tier architectures

Databases:

Oracle 19c, SQL Server, Sybase, Informix, Pervasive, DB2, Microsoft Access

Testing & Quality:

JUnit, Selenium IDE, Selenium WebDriver, SonarQube, SonarCloud, Checkmarx, Dependency Track (SBOM)

DevOps & Tooling:

GitHub, Bitbucket, Jenkins, Docker, NetBeans, Eclipse, Visual Studio Code, IntelliJ

AI-Augmented Development:

GitHub Copilot, ChatGPT, LLM-assisted architectural prototyping, AI-assisted unit test generation and edge-case analysis

PROFESSIONAL EXPERIENCE

Empower AI, Inc.

(Formerly NCI Information Systems, Inc.)

U.S. Air Force AFLCMC/SSDA Application Team

Wright Patterson AFB

Dayton, OH

Web Software Developer V | September 2020 – Present

Support and enhance the SSDA Web Security Module, a web-based security access framework used by a federal client. Perform maintenance, bug fixes, and feature enhancements across front-end, middleware, and back-end layers using Java, Spring Boot/Vaadin, PrimeFaces, JSF, JPA/Hibernate, Apache Shiro, Oracle 19c, and JBoss.

- Lead migration and modernization initiatives, including introduction of Git-based version control and repository management for legacy codebases.
- Design and implemented automated regression testing using Selenium IDE and Selenium WebDriver, integrating with Zephyr and CI workflows.
- Prototype modernization efforts to address SBOM findings, including migration from Ant to Maven and exploration of Spring Boot/Vaadin architecture.
- Develop report-generation subsystem using JasperSoft Studio.
- Integrate context-based help system using DrExplain.

AI-Augmented Development Practices:

- Incorporate generative AI tools (ChatGPT, GitHub Copilot) to accelerate prototype development and explore modernization paths.
- Use AI-assisted code scaffolding and refactoring to evaluate alternative architectural approaches during framework migration efforts.
- Leverage AI to generate test cases, identify edge conditions, and validate refactor impacts during legacy-to-modern transitions.
- Maintain full architectural ownership and manual validation in secure, compliance-driven environments.

Octo Consulting Group

U.S. Air Force AFFSO/ELF Application Team

Wright Patterson AFB

Dayton, OH

Senior Application Developer | 2019–2020

Supported ELF financial applications for the U.S. Air Force (Java, SQL Server, myBatis, Wicket). Performed maintenance, enhancements, and middleware modifications. Participated in Agile ceremonies and version-controlled workflows (Subversion, JIRA, Ant).

The Eliassen Group

General Dynamics Information Technology (GDIT)

Dayton, OH

Software Engineer / Test Automation Engineer | 2019

Designed and implemented automated test sequences for Android Tactical Assault Kit (ATAK), using TestComplete (vbScript). Developed Java-based simulation plug-ins in Android Studio. Integrated Jenkins-driven regression automation within Agile sprint cycles.

Oxford Global Resources/ Triple E Partners, Ltd./ Cohesion Corp.

Dayton/Cincinnati area

Software Engineer (Contract Roles) | 2017–2019

Delivered Java/JEE maintenance and enhancement work across automotive, financial, and retail domains.

Key contributions included:

- Middleware and web-service enhancements (Java, SQL, DB2, Oracle)
- POS system transition support and reporting subsystem updates
- Ant-to-Maven build modernization efforts
- Daily collaboration within Agile teams
- Regression and unit test validation prior to QA handoff

Technologies: Java, Spring Boot, WebLogic, IntelliJ, Maven, SQL Server, Oracle, DB2, GitHub.

Jenzabar, Inc.

Cincinnati, OH

Software Engineer (Direct Hire) | 2012–2016

Maintained and enhanced CX Student, a student and faculty-facing portion of JICS, a commercial higher-education portal application.

- Developed Java middleware plugins and server-side processes (Java, Perl, C/C++)
- Enhanced student-facing applications using JavaScript/Dojo
- Maintained Informix stored procedures and database integrations
- Coordinated with in-house and offshore teams

Technologies: Java, JBoss, Unix/Linux, Informix, n-tier architecture.

EARLIER CAREER EXPERIENCE (1990–2011)

Held progressively responsible software engineering and systems roles across federal, commercial, and industrial environments. Contributed to development, modernization, and support of client-server and early web-based enterprise systems.

Highlights:

- Designed and maintained J2EE applications (WebSphere, Struts, Hibernate, Spring, JSP) for federal government systems (2005–2011).
- Implemented PL/SQL stored procedures and enterprise reporting subsystems (JasperReports, iReports).
- Led object-oriented modernization initiatives using UML and Rational Unified Process (RUP).
- Developed and maintained Swing-based client-server applications incorporating JMS, JNDI, and MQ messaging.
- Designed multi-layered APIs in VB6/VB.NET using .NET Interop for industrial automation systems.
- Built object-oriented business rule frameworks in VB6 and C++ within 3-tier architectures.
- Early career included systems engineering, LAN administration, desktop application development, and hardware integration roles.

Technologies across this period included:

Java (J2EE), Spring, Hibernate, Struts, WebSphere, Swing, PL/SQL, Oracle 8–10g, DB2, Informix, VB6, VB.NET, C++, COM/.NET Interop, JMS, XML/XSL, UML, RUP, Solaris, Windows NT/XP.

SELECTED INDEPENDENT PROJECTS

Puzzle Gridworks (PGW)

Independent Project | 2026 – Present

Deployed Application | QuizLabs

Designed and developed an interactive visual workspace for solving, annotating, and demonstrating image-based puzzles, with an emphasis on flexible user workflows and iterative feature development.

- Built a Java/Vaadin application allowing users to import puzzle images and overlay editable text, grids, numbers, highlights, annotations, and other solving aids.
- Developed specialized puzzle-oriented tools including configurable grid objects, single-character and multi-digit entry modes, crossword numbering, sequence generation, alphabet tallies, region tools, alignment controls, and drawing/markup features.
- Implemented project persistence, export workflows, undo/redo behavior, and object-positioning features to support repeatable solving and demonstration sessions.
- Refined features through direct use on real puzzles, identifying workflow problems that were not apparent during initial implementation and converting those findings into narrowly scoped corrections and enhancements.
- Used automated testing and structured validation to reduce regressions as the application grew in scope and complexity.

AI-Augmented Workflow:

- Used conversational AI to analyze feature ideas, clarify requirements, identify edge cases, and produce implementation specifications before modifying the codebase.
 - Used GitHub Copilot as a codebase-aware implementation collaborator for focused feature additions, issue resolution, testing, and refinement.
 - Maintained architectural and product ownership by reviewing generated approaches, constraining scope, testing behavior, and rejecting technically functional solutions when they produced poor user workflows.
 - Used the project as an ongoing environment for refining a repeatable human-directed AI development process applicable to both independent and professional software work.
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The Algebrator

Independent Project | March 2025 – Present

Deployed Application | QuizLabs

Designed and deployed a Java/Vaadin algebra and puzzle-solving application that combines symbolic computation with specialized tools for numerical, logical, and word-based problem solving.

- Architected a modular Java solution integrating Symja with custom application logic and a calculator-style Vaadin interface.
- Developed specialized solver functions for integer constraints, letter-value problems, modulo relationships, digit operations, sequences, and other puzzle-oriented reasoning workflows.
- Implemented automated unit testing for arithmetic, trigonometric, symbolic, and solver behavior, including regression and round-trip validation scenarios.
- Iteratively refined application architecture, solver behavior, and UI workflows through repeated use on real problems and specification-driven testing.
- Deployed and maintained the application in a cloud-hosted environment for ongoing use and demonstration.

AI-Augmented Workflow:

- Used conversational AI for requirements analysis, feature design, edge-case review, and implementation specifications.
 - Used GitHub Copilot as a codebase-aware implementation assistant for focused feature development, debugging, testing, and refinement.
 - Retained architectural and product ownership by constraining scope, reviewing generated changes, validating behavior, and revising implementations when real use exposed design or usability issues.
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Silent E — AI-Assisted Visual Storytelling Project

Independent Multimedia Project | June 2026 – July 2026 | QuizLabs

Created a visual interpretation of Tom Lehrer’s public-domain song “Silent E,” using AI-generated imagery, editing, music, and narrative design to turn a series of word transformations into a coherent visual presentation.

- Developed visual concepts for individual word transformations such as *tub* → *tube*, *cap* → *cape*, *glob* → *globe*, and *pin* → *pine*, translating verbal humor into clear visual narratives.
- Directed iterative AI image-generation workflows, evaluating and refining generated material to maintain character, setting, visual continuity, and comedic intent across multiple scenes.
- Combined generated assets with manual selection, editing, compositing, sequencing, and presentation decisions rather than treating generated images as finished output.
- Integrated visual material with an existing musical performance, considering timing, narrative flow, and the relationship between lyrics and imagery.
- Researched public-domain and copyright considerations to ensure the project could be published responsibly.

- Completed the project as a multidisciplinary exercise combining music, visual design, AI-assisted creation, editing, and technical problem-solving.

AI-Augmented Workflow:

- Used generative AI as a directed creative tool rather than an autonomous content source, providing detailed visual requirements and repeatedly refining results that failed to meet narrative or continuity needs.
 - Applied the same iterative principles used in software development: define the desired outcome, evaluate generated output, identify deficiencies, narrow the correction, and repeat until the result meets the intended purpose.
 - Extended experience with human-in-the-loop AI beyond software development into visual communication, strengthening skills in requirements definition, evaluation, iteration, and creative problem-solving.
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Custom MIDI Pedal Controller (Arduino-Based)

Designed and implemented a custom embedded MIDI controller to translate physical pedal inputs into virtual instrument control signals.

- Designed and wired custom circuitry.
- Programmed Arduino microcontroller in C++.
- Implemented MIDI protocol handling and signal mapping.
- Debugged hardware/software interaction issues.
- Recently enhanced design using AI-assisted iteration to refine signal handling and firmware logic.

TECHNICAL EDUCATION & TRAINING

Nontraditional technical education through specialized technology programs, independent study, and long-term professional software development.

Strayer University, Newington, VA

Coursework toward Computer Information Systems (2000–2002)

Control Data Institute, Arlington, VA

Certificate in Computer Systems Maintenance (1985–1986)